



Innovative machine learning approaches for complexity in economic forecasting and SME growth: A comprehensive review

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ABSTRACT

Economic forecasting and small and medium-sized enterprises (SMEs) growth prediction have become essential tools for guiding policy, business strategy, and economic development in an increasingly data-driven world. This paper reviews recent advancements in economic regression and SME growth forecasts, with a focus on the application of machine learning (ML) techniques. Specifically, the findings highlight that the integration of ensemble methods and deep learning models has achieved significant improvements in prediction accuracy, while interpretability tools such as SHAP and LIME enhance transparency and user trust. It provides a structured analysis of diverse methodologies that includes ensemble methods, deep learning models, and interpretability tools to evaluate their effectiveness and limitations in addressing the complexities of economic and SME data. This review categorizes studies by regional focus to highlight unique challenges in different economic landscapes and the adaptability of various forecasting models. Key challenges—such as imbalanced data, feature selection, and the integration of real-time data—were identified as critical factors for enhancing prediction reliability and applicability. By comparing existing surveys and identifying gaps, this review presents actionable insights and proposes future research directions that emphasize the need for integrative models that combine Explainable Artificial Intelligence (XAI) with cross-regional data fusion for more accurate and adaptable economic forecasts. These integrative models have the potential to achieve greater regional generalizability by the offering of better decision-making tools for policymakers. The findings underscore the transformative role of ML and XAI in economic forecasting and offer valuable guidance for researchers and decision-makers to optimize forecasting models for business growth and economic planning.

Introduction

Overview of global business dynamics

Business growth and sustainability, in the modern world, are affected by numerous causes operating at local and global levels (Castillo-Vergara et al., 2024; Noman et al., 2023; Park et al., 2019; Yan et al., 2019). Globalization has provided a platform where no business, irrespective of its size or location, remains immune to international trade policies, economic downturns, and upsurges in

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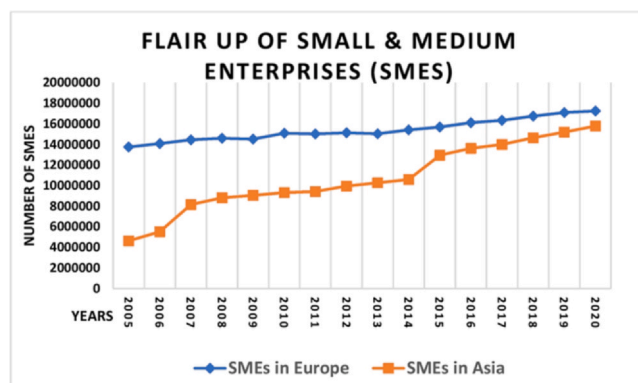


Fig. 1. SMEs growth over Europe in the last two decades (Fei et al., 2023).

technology, with changes in consumer behavior. The interconnection of business has brought in its wake the growing level of complexity in analyzing how businesses can grow, evolve, and adapt to various economic climates (Oliinyk, 2023; Štilić et al., 2023).

Global business dynamics refer to the always-changing patterns of how enterprises are established, expand, and operate within the broader economic framework (Deng et al., 2023; Khachatrian et al., 2023). Factors such as economic policies, market demands, technological innovations, and socio-political environments contribute to these dynamics. These factors, which are often volatile, require businesses to continuously adapt to remain competitive. For instance, technological innovation has drastically transformed traditional business models which allowed for the rise of digital platforms, remote work, and automated processes, which have globalized even small and medium-sized enterprises (SMEs) (Batrancea et al., 2022b; Ngo, 2023; Rumanti et al., 2023). These dynamics are essentially based on the forecasting of business growth—an art that must be possessed by every policymaker, entrepreneur, and economist. It basically informs decisions on market entry, type, and quantity of investment, but also resource allocation. On the company side, forecasting helps with making proper predictions regarding the context of market conditions and consumer preference changes. These forecasts are used by governments and institutions to shape economic policies promoting sustainable growth.

Europe represents a particularly interesting case for the study of business dynamics and has very diverse economic landscapes (Batrancea et al., 2022a; Khachatrian et al., 2023). Taking Scotland and Wales as examples, while part of the greater UK economy, these countries have shown different growth patterns given the socio-economic policies and decline in population among other things. The individual business ecosystems of each reflect the broader European trends where global economic forces interact within a set of local cultural and economic contexts (Al-Karkhi, 2024a; Batrancea, 2022; Canchari et al., 2018). Figs. 1 and 2 show the two-decade growth that SMEs in Europe have experienced, and from which a significant rise in the number and importance of SMEs can be established. The aforementioned growth brings into focus the need to grasp regional drivers and global factors that explain business thriving, in particular for countries with specific economic challenges and opportunities.

However, forecasting business growth is a challenging endeavor due to the complexity and variability of economic indicators (Huh and Kim, 2020; Wong et al., 2001). Traditional economic models often fall short in capturing the multifaceted nature of these dynamics. With the advent of advanced technologies, particularly in the fields of artificial intelligence and machine learning, there is now an opportunity to more accurately predict and analyze business growth patterns. Artificial Neural Networks (ANNs) offer a good solution by harnessing vast amounts of data to identify trends and relationships that would otherwise be difficult to detect (Al-Haddad et al., 2024, 2023; Fattah et al., 2024).

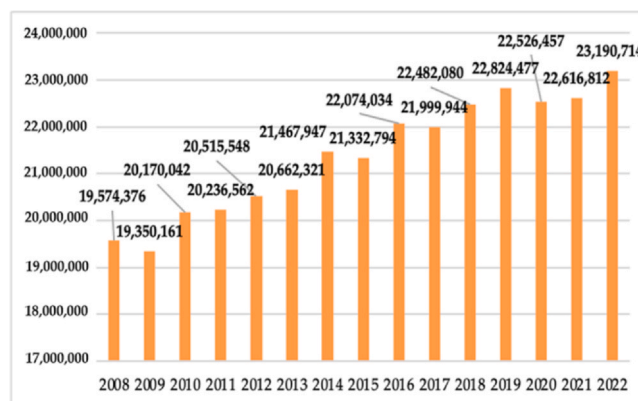


Fig. 2. Number of SMEs in the European Union (EU27) from 2008 to 2022 (Nagy et al., 2023).

Significance of business growth analysis

Perhaps one of the most critical indicators of economic health in most economies, business growth informs one about vibrancy, sustainability, and competitive ranking of economies at micro and macro levels. Knowledge on dynamics of business growth enables policymakers and economists to take decisions on strategies for economic development besides allowing businesses, investors, and stakeholders to predict likely future trends and allocate resources rightly to drive innovation. Business growth analysis goes beyond a simple numerical increase in the number of enterprises; it involves various studies of what makes an increase in the scale of enterprises take place, the sustainability of growth, and its reflections in an overall economic perspective. Business growth analysis informs one of the major aspects of economic policy and regional development strategy (Shakouri et al., 2023). In particular, various governments in transition economies rely on comprehensive data on growth to decide on the right investments in infrastructural development, fiscal policies, and trade agreements.

For instance, and as a case study, regional business growth in Scotland or Wales would mean a great deal to the overall economic state of the United Kingdom, considering that the economy is heavily dependent on regional economic policies. In fact, local support policies for SMEs, innovation hubs, and tax incentives should be well-formed in order to complement the local growth trend (Al-Karkhi, 2024b). This will be important to any policy maker bent on understanding the drivers underlying business contractions or expansions in these regions, so that adequate policies aimed at addressing regional economic disparities and thereby promoting equitable growth could be pursued.

Business growth analysis allows for a broader understanding of the processes involved in employment generation, the building up of wealth, and market competitiveness (Shakouri et al., 2023). A growing business leads to new opportunities for employment, increases demand for other goods and services, and therefore sums up the output of the general economy. This spill-over effect, or multiplier effect, states why business growth must be stimulated within an enabling environment. The pattern of employment generation from business growth helps governments and the labor market prepare the future workforce against the requirement, work on skill gaps, and ensure long-term job security. Moreover, the growth of SMEs ensures the stability of an economy, especially when economic and financial turmoil hits a country, due to their overwhelming share in the global economy; they are more adaptable and resilient against fluctuations in the local market.

But equally important in this area of analysis of business growth is how these analyses help businesses themselves contextualize their competitive positioning and future prospects in a globalized market, where rapid changes in technology and consumer preferences continuously redefine the competitive landscape. A growth trend would, therefore, enable companies to identify the upcoming market opportunities, assess the efficiency of any expansion strategy, and define resource allocation, product development, and market diversification decisions. Business growth analysis will, therefore, be required by investors who want to appreciate the long-term prospects for markets. As more investment decisions become based on predictive analytics, the ability of one to predict incidents of growth in industries and regions will make or break investment strategies pursued. With efficient analysis, investors can determine high-growth sectors where they should place capital and mitigate risks. This is to say that, to the venture capitalists and private equity firms, the gravity of understanding the growth potential of startups and SMEs using state-of-the-art technologies allows them to invest in those businesses that have the greatest potential for innovation and profitability. Business growth analysis is of particular importance in view of sustainable development. The rising consciousness of environmental and social responsibilities means, businesses increasingly will be expected to be judged not only on financial criteria but also by their contribution to sustainability. Growth compatible with sustainable development is the precondition of long-term economic viability through ensuring predictable economic performance together with tackling global challenges relating to climate change, resource depletion, as well as social injustice. Business growth analysis forms a basis for critical insight into how industries can innovate within the circle of environmental regulations and social responsibilities to help nations achieve their set targets of the Sustainable Development Goals.

Role of advanced statistics and machine learning in economic forecasting

In the contemporary landscape of economic analysis, the ability to predict future trends with accuracy is of paramount importance (Ahamed et al., 2023). Traditional statistical methods have long been employed in economic forecasting to offer insights into growth, market stability, and business expansion (Al-Karkhi, 2024a; Batrancea, 2022; Bouri et al., 2024; Norouzi, 2024; Soori et al., 2023; Zhang et al., 2024). However, with the explosion of data availability and complexity in economic ecosystems, there is an increasing demand for more sophisticated forecasting tools (Park et al., 2024a, 2024b; Williams, 2020). Advanced statistics and machine learning have emerged as powerful methods for analyzing and predicting economic dynamics, providing deeper, data-driven insights that are essential for decision-making at both micro and macro levels.

Advanced statistical techniques in economic forecasting

Advanced statistical methods evolved to master the growth of complexity and multidimensionality in modern economic data. Such methods move beyond basic linear models to include tools of time-series analysis, multivariate regression, and stochastic modeling-which uniquely interface with different facets of economic activity. Time-series analysis is one of the most important tools in economic forecasting. It enables an economist to forecast the future value of a particular variable, based on past trends, cyclical patterns, and seasonal variations. The identification of an underlying trend will, therefore, enable an economist to foresee any change in the performance of an economy and take appropriate decisions on business strategies. Regression analysis, in particular, plays a crucial role in understanding the relationships between various economic indicators. In business growth analysis, multivariate regression models can assess how factors such as GDP, interest rates, inflation, and labor force participation collectively influence

business expansion. These models allow economists to predict future business growth and understand specific drivers of this growth to target interventions and strategic decision-making.

Another helpful statistical method is that of stochastic modeling, considering uncertainty and randomness in any economic forecast. In the same vein, every stochastic model is capable of giving a subtle look at possible results of an economic event through the addition of a probability distribution and random variables. The same is much applied to the analysis of business growth, where one faces highly uncertain forecasts because of conditions in the market, change in policy, and a fluctuating world economy. The other important piece of the analysis of enterprise growth is ascertainment and calculation over time of the growth rate of enterprises. The growth rate is an approximate measure that represents the percentage increase in business activity, revenue, or market size over a specific period. It therefore gives a rather simple measure of how fast a business or sector is growing. Generally speaking, growth rates may also fluctuate from year to year due to economic cycles or external shocks; economists usually refer to the Compound Annual Growth Rate (CAGR) for a more stable measure of growth over a multiplicity of periods (Al-Karkhi, 2024a).

That being said, CAGR is still most effective when averaging out the year-over-year growth volatility and giving a more precise picture of long-term trends. It is the pace at which a business would grow if it expanded at the same rate every year over a specified period; much more indeed offers a much more consistent metric to inspect for the sustainable growth phenomenon. For instance, if a company's revenue increased from \$10 million to \$20 million within five years, the CAGR would give an average annual growth rate that led to the increase in revenue, something important for any kind of long-term strategic goal and investment decision. In economic forecasting, the application of CAGR is important not only for assessing the pace but also the sustainability of growth across multifarious regions and industries. Calculation of the annual growth rate and CAGR allows economists and business personalities to subsequently make more rational decisions on resource allocation based on market entry and expansion plans.

Machine learning

Machine learning is another sophisticated face of artificial intelligence, transforming economic forecasting with abilities to process a wide range of information and depict patterns that cannot usually be seen by conventional statistical methods (Ali et al., 2022; Baek and Arzani, 2022). At its heart, an important constituent of machine learning allows it to "learn" from data, with continuous improvement in the predictions of a model without explicit programming. It is this adaptive learning process that makes machine learning particularly useful in complex economic environments, where relationships among variables are not static but dynamic and multidimensional. One of the most impactful machine learning models in economic forecasting is the Artificial Neural Network (ANN). Modeled after the human brain's neural networks, ANNs consist of interconnected layers of nodes (or "neurons") that process data and make predictions. This layered architecture allows ANNs to handle non-linear relationships between variables, making them exceptionally suited for complex tasks like business growth forecasting.

ANNs are particularly useful when dealing with large datasets such as Open Government Data (OGD), which contains rich information on various economic indicators across regions and sectors (de Juana-Espinosa and Luján-Mora, 2020, 2019; Karamanou et al., 2023, 2022). By processing this data, ANNs can identify key trends and correlations that would be difficult to detect through traditional methods. For example, in this study, ANN models are applied to predict business dynamics across Scotland and Wales, drawing from economic data spanning 2010–2023. ANNs have the ability to integrate multiple variables—ranging from regional policy impacts to market fluctuations—provides a more robust and reliable forecast of business growth.

The flexibility of machine learning models, particularly ANNs, also allows for continuous refinement. As more data becomes available or as economic conditions evolve, the model can be retrained to adjust its predictions accordingly. This dynamic capability is invaluable in economic forecasting, where unexpected events (such as global pandemics or financial crises) can rapidly alter market conditions. Furthermore, ANNs offer predictive accuracy far beyond traditional methods, as seen in this study's ability to achieve high levels of accuracy with Mean Absolute Percentage Error (MAPE) values as low as 0.8 % and coefficients of determination (R^2) reaching 0.97. In addition to ANNs, other machine learning techniques, such as decision trees, support vector machines (SVM), and ensemble learning methods, have proven effective in economic forecasting. These techniques, when applied alongside advanced statistical methods, provide a comprehensive toolkit for analyzing and predicting business growth.

This translates to the fact that such an integration of advanced statistical techniques and machine learning in economic forecasting opens up new unexplored vistas for predictive accuracy and adaptability. While advanced statistics provide a very strong foundation in understanding interrelations among economic variables, it is machine learning that extends this very foundation by coping more effectively with complexity and large-scale data. Consequently, the interaction between the two methods allows for properly developing more accurate and reliable actionable insights into the dynamics of business growth, thus enabling prudent decision-making on the part of policy and business leaders and investors.

Introduced novelties

This review provides several key contributions to the field of SME and economic forecasting by integrating machine learning methodologies and addressing specific gaps identified in prior research. The novelties introduced in this review include:

By synthesizing a broad spectrum of machine learning techniques—such as ensemble methods, deep learning, and interpretability approaches—this review systematically compares the strengths and limitations of each, offering a clear understanding of their applications in economic and SME forecasting.

Analysis of regional studies across diverse economic landscapes. By categorizing methodologies according to regions, the review highlights specific geographical challenges and adaptability factors, aiming to improve cross-regional forecasting models.

This review uniquely emphasizes handling imbalanced data, real-time data fusion, and feature selection, which are often overlooked but critical for reliable SME predictions. Our analysis brings these challenges to the forefront, paving the way for more accurate and adaptable forecasting models.

Unlike traditional reviews, this work highlights the importance of model interpretability and transparency, offering insights into how interpretability techniques can make machine learning models more actionable for policymakers and business leaders.

Finally, the review proposes a pathway for future research, advocating for integrative approaches that combine real-time data, multi-source fusion, and explainable AI to create more robust and scalable forecasting frameworks.

The remainder of this paper is structured as follows: [Section 2](#) provides a detailed Literature Survey. [Section 3](#) focuses on Machine Learning Techniques in Economic Forecasting and examines the most impactful algorithms. [Section 4](#) identifies Research Gaps and Positioning which highlights underexplored areas and future directions. Finally, [Section 5](#) offers a Conclusion to summarize the review's contributions and suggesting pathways for advancing economic forecasting model.

Literature surveys

Introduction to literature review

The dynamic and multifaceted nature of business growth has made its forecasting a critical area of economic research. As economies across the globe continue to grow more interconnected, the factors influencing business development and expansion have evolved in both scale and complexity. Predicting business growth requires not only a deep understanding of economic indicators but also the ability to process and analyze data with unprecedented volume and speed. Advanced statistical techniques and ML models have made business forecasting methods increasingly enhanced in their accuracy and applicability in recent decades. Such tools allow one to gain insight not possible with traditional econometric models, and thus one acquires valuable and actionable intelligence on economic policy and corporate strategy for various stakeholders.

In this context, the literature has broadly examined the foundational elements of business forecasting, from traditional approaches that utilize linear models to contemporary methods based on artificial intelligence (AI). While most research traditionally uses time-series analysis and multivariate regression to spot trends and relationships in economic data, those methods have limited capabilities to rightly model nonlinear and interdependent relationships common to many types of modern economic data. This has created a shift toward machine learning approaches-especially ANNs and other data-driven algorithms-which enable researchers to make more realistic forecasts of business growth in highly complex and volatile economic environments. Moreover, there is a whole change in the face of economic research with the emergence of OGD. First, by providing high-quality open data to researchers, OGD has made analyses more transparent and extensive on regional and national business dynamics. Multiple regional governments and organizations have published some datasets that bear relevance to the scope of various economic indicators, which are now accessible for conducting comprehensive, high-resolution analysis at the macro scale. While OGD gains momentum to become an indispensable part of economic research, the amalgamation with machine learning models opens new avenues for real-time forecasting and examination of business growth trends, enabling responsive and more informed decision-making.

It is, therefore, expected that this chapter will build a critical understanding of the available literature on the forecasting of business growth, taking into consideration both the previously established methods and recent developments. The review first gives an overview of some key economic indicators and their relations to business performance forecasting. Second, this review discusses the implications of OGD for economic analysis, particularly on the availability of quality datasets, that may be used to train machine learning algorithms. The next section of the chapter reviews previous literature reviews on economic forecasting, identifying the trends and developments this area of consensus has undergone through the years. Lastly, the contribution of machine learning to the improvement of economic forecasting was discussed in the chapter, identifying the potential role of ANNs and other more advanced models in overcoming the shortcomings of traditional approaches.

Business growth forecasting and economic indicators

Accurately predicting business growth can be important for economic planning as it helps policymakers and enterprises make informed decisions using forecasted market developments. Traditional approaches to forecasting rely heavily on leading economic indicators such as Gross Domestic Product, which is a macroeconomic indicator of growth, employment levels, consumer expenditure, and investment levels. However, recent advances proved that alternative data sources and more granular economic indicators can be used in enhancing the predictive power of preliminary estimates, further refining growth forecasts, and expanding higher accuracy to a wide range of economic contexts.

One innovative approach to forecasting involves the use of news-based indicators. [Thorsrud \(2020\)](#) ([Thorsrud, 2020](#)) proposed a model that integrates textual information from daily business newspapers to construct a daily business cycle index. This model has succeeded in embedding high-frequency economic fluctuations and has improved the no-arbitrage GDP growth forecast based on a dynamic factor model that imposes sparsity in factor loadings. News really shows how critical and unstructured textual data might actually help reduce noise for many traditional indicators and provide much more detail with respect to the business cycle, one that would make so much better correlation with real-time economic conditions. A critical perspective involved in business growth forecasting is that of assessing business uncertainty. [Altig et al. \(2022\)](#) ([Altig et al., 2022](#)) developed a novel survey mechanism called the Survey of Business Uncertainty (SBU), which captures firm-level subjective probability distributions over anticipated outcomes

like sales growth and employment. This survey provides a nuanced view of business expectations and uncertainty, finding that high subjective uncertainty correlates with larger forecast errors and revisions. The SBU combines these firm-level distributions into indices of business expectations and uncertainty that accurately capture shifts in growth rates and prove quite useful as a leading indicator for policymakers concerned with the private sector.

Econometry has also been closely studied in the context of unexpected global shocks, such as the COVID-19 pandemic. Altig et al. (2020) (Altig et al., 2020) examined various economic uncertainty indicators which include implied stock market volatility, policy uncertainty in news media, and social media discussions on economic conditions. These indicators blew out to record levels during the pandemic, while these variations in industrial production directly followed uncertainty shocks. The findings of this study put into perspective the relevance of real-time monitoring of economic indicators, especially in periods of crisis, for a better comprehension of the underlying effects on business growth and industrial output.

Adding up, economy must consider the influence of large economic shocks on specific industries. For instance, Gudmundsson, Cattaneo, and Redondi (Gudmundsson et al., (2021) (2021) analyzed the global air transport industry during COVID-19 and estimated that passenger demand recovery would vary widely across regions. This study has been able to clearly outline such significant shocks on the issue of growth and time of recovery, thereby showing that regional disparities in economic resilience should form the basis of industry-specific forecasts. Industry-wise, this study was able to draw the fact that there are sectors like air transport which, though having different recovery paths, impact regional and global economic stability, and hence this should be an important reference for other similar industries' business growth forecasts.

Technological advancements in Information and Communication Technology (ICT), are increasingly recognized as drivers of economic growth. Vu, Hanafizadeh, and Bohlin (2020) (Vu et al., 2020) conducted a comprehensive literature survey exploring ICT's role in economic development, which reveals the positive correlation between digital infrastructure and business growth. Their findings suggest a shift in focus toward understanding how new digital technologies influence economic performance, advocating for deeper studies into the mechanisms by which ICT advances drive productivity and sustainable growth. This research aligns with emerging priorities in business forecasting, as technology adoption becomes an integral factor in predicting growth dynamics.

Finally, the digital maturity of SMEs has emerged as a critical economic indicator, especially within the European Union. Brodny and Tutak (2022) (Brodny and Tutak, 2022) evaluated SMEs across EU-27 countries to assess their levels of digital adoption and identified significant disparities in digital maturity between regions. The present study, by applying entropy-based methods and principal component analysis, underlined the relation of digital readiness with economic parameters and showed that usually, the higher the digital maturity, the better the rate in business growth. These insights into the trends of digitalization create the basis for integrating digital adoption metrics within forecasting models, at least in as far as businesses continue to embrace Industry 4.0 technologies. To put it in one word, the forecasting of business growth has elevated traditional economic indicators to a new dimension that comprises an array of data sources and state-of-the-art methodologies. As crystal clear from several studies, textual analysis, uncertainty measures, technology indicators, and industry-specific information broaden the forecast by offering a grander view of prospective business growth. These changes signal the rising demand for multi-dimensional, comprehensive models that incorporate new data types and advanced statistical techniques to accurately forecast growth in today's more complex economic environment.

Existing surveys in economic forecasting and SMEs growth

To capture the breadth of recent studies in economic forecasting and business growth, this section provides a structured overview of nine recent literature reviews. These reviews cover various methodologies, applications, and challenges in areas related to machine learning, artificial intelligence, SMEs, and broader economic forecasting. The tables below synthesize the main themes, methodological insights, and future research recommendations from these reviews, offering a comprehensive snapshot of the current landscape in business growth prediction and forecasting methods. Table 1 presents a high-level summary of the primary focus and methodological approaches used across the reviews. This table categorizes each article based on its thematic focus—such as SME growth, default prediction, and machine learning applications—and the main methodologies discussed with a highlight on the evolution of research techniques in economic forecasting and machine learning. Table 2 further categorizes the reviews by their identified challenges, contributions to the field, and recommendations for future research. This table underscores the research gaps that exist across these fields and also points to emerging trends and necessary advancements in methodology and data utilization.

Diverse drivers of economic forecasting are revealed, summarized in Table 1. Each review has shed light on machine learning and artificial intelligence applications across industries. Reviews of SME-specific challenges, such as default prediction and technology adoption, point to the emergent use of AI in improving predictive accuracy and operational efficiency. Notably, topics such as default prediction and IoT applications in SMEs were investigated using statistical methods and machine learning to illustrate the interdisciplinary approach of combining finance, business, and data science in order to improve predictive accuracy.

Table 2 summarizes the general challenges and future research directions put forward in the reviews. Note that issues related to data quality, validation techniques, and computational demands for nonlinear forecasting models keep recurring in light of the sensitivity of machine learning applications to varied economic forecasting domains. Common suggestions for future work include wider availability of AI and IoT in general, more extensive empirical comparisons of the methods, and special efforts to cater to the needs of resource-constrained sectors, such as SMEs. These collective findings suggest that, while machine learning and AI offer promising advances, challenges remain in achieving consistent, high-quality predictions in real-world applications.

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Table 1
Overview of themes and methodologies in recent literature reviews.

Ref.	Primary Focus	Methodologies Discussed	Sector/Context
(Beliaeva et al., 2023)	AI in predicting SME growth	Comparison of AI and traditional growth methods	SME Entrepreneurship
(Cheraghali and Molnár, 2024)	SME default prediction	Estimation, re-balancing, feature selection, validation methods	Business Finance
(Gao et al., 2024)	Machine learning in business & finance	Deep learning, neural networks, predictive analytics	Business & Finance
(Ahmad and Chen, 2020)	Machine learning in energy forecasting	Supervised ML algorithms, Bayesian neural networks	Energy Systems
(Ciampi et al., 2021)	Default prediction for SMEs	Statistical analysis, bibliometric methods	Finance, Management, Accounting
(Cordes et al., 2022)	Chebyshev polynomials for economic forecasting	Non-linear approximation, neural networks	Low-income economic forecasting
(Pramanik and Jana, 2023)	ML research trends in business	Topic modeling, latent Dirichlet allocation	Business Operations, Marketing
(Vajjhala, 2024)	ML for SMEs	Exploratory review, cost-effective ML applications	SMEs across various industries
(Hansen and Bøgh, 2021)	AI and IoT in SMEs	Survey analysis, predictive analytics capabilities	Manufacturing SMEs

Table 2
Challenges, contributions, and future directions in surveys.

Ref.	Key Challenges Identified	Main Contributions	Future Research Recommendations
(Beliaeva et al., 2023)	Low predictive power in SME growth models	Novel use of AI for growth prediction	Broaden AI application in entrepreneurship
(Cheraghali and Molnár, 2024)	Lack of validation methods in default prediction	Comprehensive methodology review	Empirical comparison of estimation methods
(Gao et al., 2024)	Data quality, interpretability	Overview of ML applications in finance	Enhance ML for cryptocurrency, crime detection
(Ahmad and Chen, 2020)	Real-time data integration	Comparison of energy forecasting models	Real-time forecasting in energy applications
(Ciampi et al., 2021)	New challenges from economic crises	Systematic review of SME predictors	AI and macro-data integration
(Cordes et al., 2022)	Computational expense of non-linear models	Review of Chebyshev polynomial applications	Sustainable development models for low-income
(Pramanik and Jana, 2023)	Gaps in HR and strategy ML adoption	Trends in ML across business domains	Increased ML adoption in HR and strategy
(Vajjhala, 2024)	Limited resources in SMEs	Exploratory review of ML applications	ML for decision-making in resource-limited SMEs
(Hansen and Bøgh, 2021)	Access to AI and IoT technologies	Survey of AI/IoT in SMEs	Broaden IoT access in manufacturing SME

wider availability of AI and IoT in general, more extensive empirical comparisons of the methods, and special efforts to cater to the needs of resource-constrained sectors, such as SMEs. These collective findings suggest that, while machine learning and AI offer promising advances, challenges remain in achieving consistent, high-quality predictions in real-world applications. Figs. 3 and 4 illustrate the distribution of topics and challenges across recent literature reviews in economic forecasting. Fig. 3 shows that "SME Applications" holds the highest representation as it highlights the emphasis on AI and ML applications for SMEs. Meanwhile, Fig. 4 identifies "Data Quality & Validation Issues" as the most frequently cited challenge and underscores ongoing concerns about the reliability and processing of data used in predictive modeling. These insights reveal priority areas in both focus and methodological improvement for future research in economic forecasting.

Machine learning in economic forecasting

ML applications in SME and economic forecasting have seen a variety of methodological approaches, each tailored to address specific challenges within SME contexts, including financing, credit risk, crisis management, and performance prediction. Recent

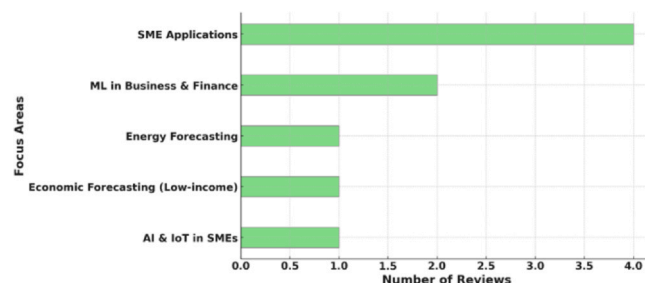


Fig. 3. Broader focus areas in surveys on economic forecasting.

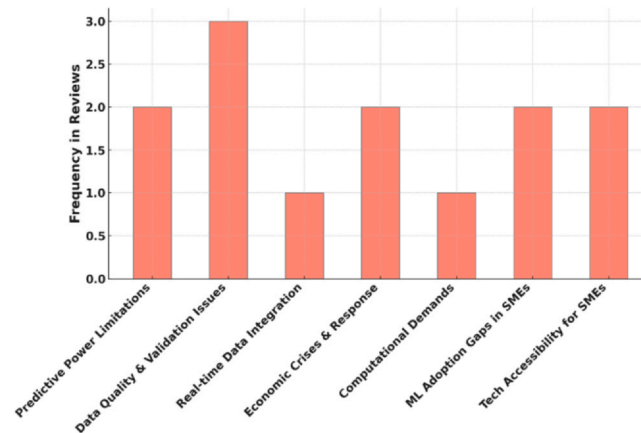


Fig. 4. Challenges in economic forecasting literature reviews.

developments signal interest in ensemble and hybrid methods to get high-accuracy and reliable outputs for different forecast needs. Based on these, Ma et al. combined an improved extreme learning machine for risk prediction in SME financing, resulting in a significant reduction of Type-2 error by 70.24%. Similarly, Zhu et al. presented the efficacy of RS-MultiBoosting for credit risk prediction in supply chain finance, focusing on traditional metrics and SCF-specific factors for boosting accuracy. Daoud and Kammoun used ML-based forecasting to identify eight critical factors affecting the adoption of e-commerce in SMEs; from this, they proved that ML outperformed logistic regression in predicting these drivers. Wang et al. employed cost-sensitive random forests and multi-source information fusion, finding that the financial indicators are highly predictive for credit risk in SCF, while again emphasizing the need to treat the issue of handling imbalanced data. Table 3 summarized these methodologies and presented the effectiveness of those in SME and economic forecasting.

For SME failure prediction, Schalck and Yankol-Schalck found XGBoost to be the most accurate, and SHAP interpretability techniques revealed that both financial and nonfinancial variables are significant failure factors. Crisis management was tackled by Zhao et al., who applied a stacking fusion model incorporating logistic regression, backpropagation neural networks, and XGBoost, achieving 85.8% accuracy in crisis prediction with superior performance from XGBoost. Jain et al. explored SME performance prediction using recurrent neural networks (RNN) and gradient boosting, demonstrating the model's high accuracy in assessing strategic planning impacts on performance. In manufacturing SCF, Xia et al. used random forest models to analyze credit risk and

Table 3

Methodological approaches and key findings in machine learning applications for SME and economic forecasting.

Ref.	Focus Area	ML Technique(s) Used	Key Findings
(Ma et al., 2023)	Financing risk in SMEs	Improved Extreme Learning Machine	Reduced Type-2 error by 70.24%, reliable for financing risk forecasts
(Zhu et al., 2019)	Credit risk in supply chain finance	Hybrid Ensemble (RS-MultiBoosting)	Enhanced accuracy for SCF credit risk, emphasizing traditional and SCF-specific factors
(Daoud and Kammoun, 2024)	E-commerce adoption in SMEs	ML-based forecasting	Identified 8 critical adoption factors, higher accuracy than logistic regression
(Wang et al., 2023)	Credit risk in SCF	Cost-sensitive Random Forest, multi-source info fusion	Financial indicators are most predictive, importance of imbalanced data handling
(Schalck and Yankol-Schalck, 2021)	SME failure prediction	Probit, Lasso, XGBoost	XGBoost outperforms; SHAP shows significant influence of financial and nonfinancial factors
(Zhao et al., 2023)	Crisis management in SMEs	Stacking fusion of LR, BPNN, XGBoost	85.8% accuracy for crisis prediction, XGBoost outperforms traditional methods
(Jain et al., 2018)	Performance prediction of SMEs	Recurrent Neural Networks, Gradient Boosting	Accurate predictions of strategic planning's impact on SME performance
(Xia et al., 2023)	SME credit risk in manufacturing	Random Forest with partial dependency	Financial and operational profiles crucial; Random Forest shows high predictive power
(Wang and Zhang, 2024)	SME innovation and regional economic growth	XGBoost, IoT integration	High accuracy in coupling coordination, IoT data enhances insights
(Boubaker et al., 2023)	Performance of MSMEs	Hybrid DEA and ML (Tobit, Lasso, RF)	Hybrid approach improves efficiency, allows ranking and predictive performance
(Lu et al., 2024)	Revenue and employment growth of SMEs	BERT-TCN (text and sequential data)	Transformer networks handle complex, bilingual datasets, improving feature learning
(Zhu et al., 2016)	SME credit risk in SCF	Ensemble (RS-RAB)	RS-RAB shows superior performance for quarterly credit risk prediction
(Yoon, 2021)	Real GDP growth	Gradient Boosting, Random Forest	Gradient Boosting achieves more accurate forecasts than IMF/BOJ benchmarks

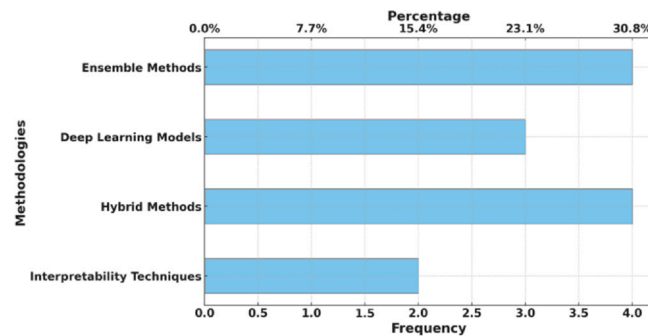


Fig. 5. Methodological approaches in SME and economic forecasting studies.

found financial and operational profiles to be highly predictive, with partial dependency analysis enhancing the model's interpretability. Wang and Zhang leveraged IoT data with XGBoost to assess SME innovation and regional economic growth and demonstrated high accuracy in coupling coordination analysis, with IoT integration adding value.

Boubaker et al. applied a hybrid DEA and machine learning approach, using models like Tobit and random forest to predict the performance of micro, small, and medium enterprises (MSMEs). This hybrid approach allowed for efficiency improvements and ranking comparisons, particularly for Vietnamese MSMEs. In another innovative application, Lu et al. combined BERT and temporal convolutional networks (TCN) to manage bilingual, structured, and sequential datasets in forecasting SME revenue and employment growth, improving feature learning and addressing class imbalance challenges. For credit risk prediction in SCF, Zhu et al. found the RS-RAB ensemble model effective in quarterly predictions, outperforming traditional methods. Lastly, Yoon compared gradient boosting and random forest for Japan's GDP growth prediction, showing gradient boosting to be more accurate than International Monetary Fund (IMF) and Bank of Japan (BOJ) benchmarks. This variety of methodologies emphasizes ensemble methods (e.g., RS-MultiBoosting, RS-RAB), deep learning (e.g., BERT, XGBoost), hybrid frameworks, and interpretability techniques like SHAP to support decisions across multiple domains. Fig. 5 illustrates the methodological landscape, where ensemble and hybrid methods dominate with 30.8% each, followed by deep learning models (23.1%) and interpretability tools (15.4%) as detailed in Table 4. These findings highlight the strategic importance of ensemble and hybrid approaches and especially if applied to complex SME scenarios and economic forecasting.

The application of machine learning models to SME and economic forecasting involves addressing specific challenges, integrating innovative approaches, and setting directions for future research. Key challenges, such as imbalanced data, effective feature selection, and ensuring model generalizability, are frequently encountered in forecasting models for SMEs. To tackle these, recent studies have introduced innovative techniques that include the Grey Wolf optimization algorithm for intelligent feature selection in enterprise financing. Additionally, methodologies like cost-sensitive random forests and random subspace-based ensemble methods address imbalanced data, enhancing the accuracy of credit risk predictions in SMEs. Tables 5 and 6 show a detailed and frequency-based analysis of this case, respectively. Fig. 6 visually demonstrates the frequency and distribution of these challenges and innovations, with imbalanced data and model generalizability appearing as the most prominent concerns.

Feature selection stands out as a significant area of focus, contributing to efficient data preprocessing and accurate predictions. Innovations such as RS-MultiBoosting and Grey Wolf optimization help refine feature selection processes and yielded higher accuracy in scenarios like SCF and SME failure prediction. The application of these methods ensures that critical variables, whether in financing or supply chain contexts, are prioritized for more robust forecasting outcomes. Moreover, the use of advanced interpretability techniques, such as SHAP values, provided good insight into variable importance and enhanced model transparency and offering decision-makers a clearer understanding of the key drivers behind predictions.

Looking ahead, future research directions suggest broadening model applicability across different regional and industrial contexts and emphasizes data fusion and real-time data applications as Table 6 enlists. Model scalability and adaptability to dynamic environments are also emphasized, particularly through coupling analysis, IoT integration, and deep learning applications like BERT and TCN in handling complex data structures. These advancements demonstrate a forward-looking approach in SME and economic forecasting, with ongoing efforts to refine ML models and enhance predictive accuracy in increasingly complex financial landscapes.

The application of machine learning to economic and SME forecasting exhibits a diverse range of regional focuses as it reflects both the global relevance of predictive analytics and specific challenges faced by different economies. Studies have prominently

Table 4
Comparative analysis of AI approaches.

Category	Frequency	Percentage (%)
Ensemble Methods (e.g., RS-MultiBoosting, RS-RAB)	4	30.8 %
Deep Learning Models (e.g., BERT, TCN, XGBoost)	3	23.1 %
Hybrid Methods (e.g., Hybrid DEA-ML, Stacking Fusion)	4	30.8 %
Interpretability Techniques (e.g., SHAP)	2	15.4 %

Table 5
Challenges, innovations, and future in SME and economic forecasting models.

Ref.	Key Challenges Addressed	Innovations Introduced	Future Directions
(Ma et al., 2023)	Imbalanced data, prescreening risk indicators	Grey Wolf optimization for feature selection	Enhance ELM for cross-domain applicability
(Zhu et al., 2019)	Small sample sizes	RS-MultiBoosting ensemble method	Apply SCF model across broader SME sectors
(Daoud and Kammoun, 2024)	Limited insights into e-commerce drivers	Application of ML to TOE framework	Explore deeper cross-country studies on EC adoption
(Wang et al., 2023)	Imbalanced classes in credit risk	Cost-sensitive Random Forest	Extend multi-source fusion to different financial domains
(Schalck and Yankol-Schalck, 2021)	Failure prediction accuracy in SMEs	SHAP values for interpretability of XGBoost	Use multi-variable datasets with different regional conditions
(Zhao et al., 2023)	Crisis latent period recognition	Stacking fusion to improve accuracy	Enhance model to capture sector-specific crises
(Jain et al., 2018)	Data processing challenges for SMEs	4-module performance prediction framework	Refine framework for larger datasets and dynamic environments
(Xia et al., 2023)	Real-time credit risk assessment	Partial dependency plot for key variables	Broaden dataset for other manufacturing-focused SME evaluations
(Wang and Zhang, 2024)	Coupling of SME growth with regional prosperity	IoT and XGBoost integration	Apply coupling degree analysis to different regional datasets
(Boubaker et al., 2023)	Non-convexity in DEA for MSMEs	Hybrid DEA with econometric and ML techniques	Extend to cross-national MSME performance analysis
(Lu et al., 2024)	Feature extraction from complex, bilingual data	BERT-TCN for text and sequence handling	Explore Transformer applications in other business datasets
(Zhu et al., 2016)	Accurate SCF credit risk assessment	Random Subspace-Real AdaBoost ensemble	Compare with other ensemble methods for similar applications
(Yoon, 2021)	Hyperparameter optimization for macroeconomic data	Cross-validated Gradient Boosting	Apply cross-validation to other forecasting models

Table 6
Summary of challenges, innovations, and future in SME and economic forecasting.

Challenge/Innovation	Frequency	Percentage (%)
Imbalanced Data	4	30.8 %
Feature Selection	3	23.1 %
Data Fusion & Real-Time Data	2	15.4 %
Model Generalization & Applicability	4	30.8 %

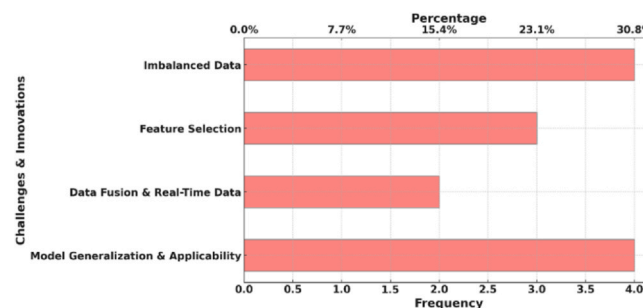


Fig. 6. Challenges and innovations in machine learning applications for SMEs.

concentrated on Asian countries, particularly China, where research in SME financing and credit risk within the supply chain finance (SCF) domain has been active as Table 7 shows. This emphasis on China demonstrates a strong interest in using machine learning to address economic volatility and financing risks among SMEs within an expanding market economy. Similarly, emerging economies, such as Tunisia and Vietnam, have been the focus of studies on e-commerce adoption and MSME performance, respectively which showcases a need for tailored ML approaches that address region-specific economic conditions, again as Table 7 enlists.

Research extending beyond single-country contexts includes crisis management frameworks applied globally and performance prediction models focused on developing nations. These studies highlight a broader interest in enhancing SME resilience and economic forecasting across regions facing unique challenges. As shown in Fig. 7, Asia leads in regional focus, accounting for a substantial share of the research efforts, followed by Europe and other regions with smaller, yet impactful, contributions. The North

Table 7
Regional focus in machine learning applications for economic and SME forecasting.

Ref.	Primary Region	Focus on SMEs / Economic Forecasting
(Ma et al., 2023)	China	SME Financing Risk
(Zhu et al., 2019)	China	SME Credit Risk in SCF
(Daoud and Kammoun, 2024)	Tunisia	E-Commerce Adoption in SMEs
(Wang et al., 2023)	China	Credit Risk in SCF
(Schalck and Yankol-Schalck, 2021)	France	SME Failure Prediction
(Zhao et al., 2023)	General (Global)	Crisis Management in SMEs
(Jain et al., 2018)	Developing Countries	Performance Prediction in SMEs
(Xia et al., 2023)	China	Manufacturing SME Credit Risk in SCF
(Wang and Zhang, 2024)	China (Yangtze River Economic Belt)	Regional Economic Growth and SME Innovation
(Boubaker et al., 2023)	Vietnam	MSME Performance
(Lu et al., 2024)	Canada	Revenue and Employment Growth Prediction
(Zhu et al., 2016)	China	SME Credit Risk in SCF
(Yoon, 2021)	Japan	Real GDP Growth Forecasting

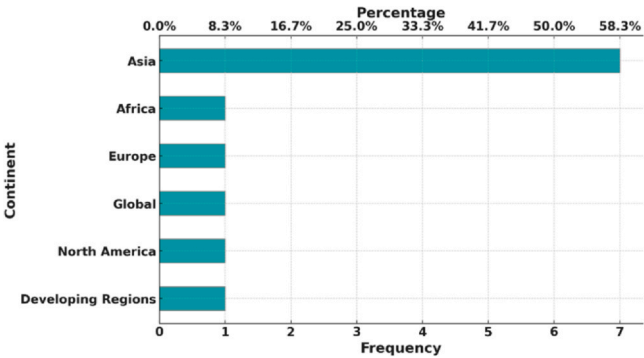


Fig. 7. Regional focus of SME and economic forecasting research by continent.

American studies, such as those in Canada, emphasize advanced ML models for revenue and employment growth, while Japan focuses on real GDP growth as it illustrated a more macroeconomic perspective. Finally, the frequency and distribution of regional focuses underscore the widespread applicability of machine learning in various economic environments, with particular attention to financing and risk prediction. As reflected in Table 8, this geographical diversity suggests a promising avenue for cross-regional studies, which could further enhance the generalizability and adaptability of machine learning models across distinct economic and cultural contexts.

Research gaps and positioning

ML approaches to economic and SME forecasting are in a stage of extensive development. However, large research gaps restrain the effectiveness and adaptability of current methods and their broader application. The main challenge is how to deal with imbalanced data; in financial forecasting of SMEs, asymmetrical data distribution impairs the reliability of predictions. While some methods try to alleviate these issues through cost-sensitive random forests, ensemble techniques like RS-MultiBoosting and RS-RAB, and advanced algorithms such as Grey Wolf Optimization for feature selection, most of the current models lack robustness against highly imbalanced datasets across varied contexts. While the high Type-2 error reductions obtained in applying such frameworks in studies are indicative of promise, this remains an area where universally adaptable solutions are still being looked for to ensure dependable predictions of financing and credit risk across diverse SME ecosystems.

Another critical gap has been the regional specificity and limited scalability of the machine learning models. Research clearly focuses on specific countries, such as works that have focused on China, France, and regions within developing economies. This trend

Table 8
Regional focus frequency and percentage.

Challenge/Innovation	Frequency	Percentage (%)
Imbalanced Data	4	30.8 %
Feature Selection	3	23.1 %
Data Fusion & Real-Time Data	2	15.4 %
Model Generalization & Applicability	4	30.8 %

is indicative of a regional bias that constrains the findings on adaptability to global or cross-regional contexts. While local studies have their merit, a lack of persistence across other economic environments creates a gap in the understanding of how such models potentially function across different regulatory landscapes, economic structures, and characteristics of SMEs. By expanding studies beyond being single-country or limited-region applications, it allows the model generalization and gives way to comparisons that capture diverse influences of regulations, structures, and economies. This is relevant for regions such as Europe, where regulations on a country-specific level, combined with unique market dynamics, affect the ranking of SMEs differently than in other regions.

While promising innovations like IoT integrations, deep learning, and hybrid ensemble methods are foreseen, real-time data integration and multi-source data fusion are not well exploited. Working applications of real-time IoT data were achieved, for example, in those studies targeting supply chain and manufacturing SMEs, where XGBoost and other machine learning methods ensured promising performance in real-time risk assessment and performance evaluation. However, data from multiple sources is rarely integrated into structured financial data, such as textual economic reports or IoT data streams, within a forecasting model. Integration of these types of data may yield more dynamic and accurate models, especially in turbulent economic times where the possibility of real-time adjustments would make those forecasts more actionable. This is a very critical gap in forecast analytics, as timely and agile predictions are highly sought after by decision-makers whose operations depend on changing market conditions.

Besides these technical and regional deficits, interpretability and transparency are also lagging behind in the current forecasting models. Although SHAP values have represented variable importance, enabling the stakeholders to understand the forecasts made, most studies still consider interpretability as a feature that can be used but is not necessary. Because conventionally, this is not emphasized, it limits how much value can be extracted from ML-based forecasts by especially policymakers and business leaders looking for transparency and actionable insights that could drive their strategic decisions. Models imbued with interpretability at their core architecture might prove the dual advantage in giving robust, data-driven predictions while enhancing confidence in model recommendations among non-technical stakeholders.

Based on this, the present thesis positions itself at these hiatuses; it articulates a general machine learning framework to be designed for issues related to imbalanced data, regional adaptability, real-time data fusion, and increased model transparency. It will, in particular, focus on Europe, where such adaptability is important because of diverse regulations and market structures. The research will develop a model underlining cross-regional applicability, driven by economic and structural nuances across European economies. By integrating ensemble methods and techniques of real-time data integration, the model envisages dynamic predictions based on real economic contexts, hence addressing one of the main gaps in the existing forecasting literature. Besides this, by embedding interpretability techniques, such as SHAP, as an integral part of the model, this thesis also advances its accessibility and applicability for academic and practical use. Therefore, this study alone justifies its contribution to the literature of economic forecasting by providing a well-scaled, adaptable, and most definitely more transparent model in light of these identified gaps that will enable a higher amount of informed decision-making across a wide range of economic landscapes.

Conclusion

In conclusion, this paper provided a comprehensive review of recent advancements in economic forecasting and SME growth prediction, with a particular focus on machine learning applications. The study synthesized a broad array of methodologies, highlighting strengths and limitations across ensemble methods, deep learning models, and interpretability techniques. Through an analysis of regional studies, it identified specific challenges and the adaptability of various approaches in different economic contexts. Additionally, the review underscored key issues such as imbalanced data, real-time data fusion, and the need for robust feature selection, which are essential for improving the reliability of predictive models in SME and economic contexts. By examining gaps within the existing literature, this review offered future research directions, advocating for integrative models that incorporate real-time data, cross-regional adaptability, and explainable AI to enhance forecasting accuracy and applicability. The findings underscore the potential of machine learning to transform economic forecasting, providing valuable insights for researchers, policymakers, and practitioners aiming to refine prediction strategies for economic growth and SME development.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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